National Institute of Technology Patna

Department of Computer Science and Engineering

Mid Sem Examination – August – December 2024

Course Name: Programming in C | Course Code: CS14102 / CS17101

Time Allowed: 2 Hours | Total Marks: 30

Department: CSE / Mathematics and Computing | Semester: 1st

Instructions:

- **This Question Paper contains two sections. CSE3 & Mathematics and Computing should attempt Section I only. CSE1 and CSE2 students should attempt section II only**
- **Assume missing data if any.**

Section I

(For CSE3 & Mathematics and Computing Students only)

Q1. [3+2+5=10 Points][CO1,3,4][BL3]

a. Write a C program that utilizes a two-dimensional array. The program should prompt the user to input p number of rows, with each row containing q elements. After receiving the input, the program should display the elements in the output.

b. Write the C program to swap number using the third variable.

c. Write a C program that accepts three integers from the user and determines which of these integers is the largest. The program should implement input validation to ensure that only valid integer values are accepted. Draw The flow chart for the same.

Q2. [5+5=10 Points][CO3][BL2]

a. How switch statement can be effectively used to represent and handle different months of the years by assigning unique integer value to each month. Write the C program for the same.

b. Write any program using **break** and **continue** statement? Explain when you will use break and continue statements according to your program.

Q3. [5 Points][CO3][BL3] Suppose in your college institute library charges the fine for every book returned late. For the first five days the fine is one rupee, for 6 to 10 days fine is two rupees and above 10 days fine is 10 rupees. Suppose if you return book after 30 days your membership will be cancelled. Write a C program to accept the number of days the member is late to return the book and display the fine or the appropriate message.

Q4. [1x5=5 Points][CO2,3][BL3]Will the following code snippet compile, if so, what will be output. Explain your answer in all cases:

a.
```
main()
{ int a=100, b, c;
if(a>=200)
b=30;
c=20;
printf("%d %d",b,c); }
```

b.
```
main()
{ int a=10, b=20;
if(a==b)
printf("%d %d",a,b);
}
```

c.
```
main()
{ int x=10; float y=10.0;
if(x==y)
printf("x and y are equal ");
else
printf("x and y are not equal"); }
```

d.
```
main()
{ int x=66;
char y='B';
if(x==y)
printf(" x is equal to y");
else
printf(" x is not equal to y); }
```

e.
```
main()
{ int i=4,j=-1,k=0,w,x,y,z;
w=i || j || k
x= i && j && k
y= i && j ||  k
z= i ||  j && k
printf("w=%d, x=%d, y=%d, z=%d" ,w,x,y,z); }
```

Please turn over for Section II

Section II
(For CSE1 and CSE2 students only)

Q1. **[7+3=10 Points][CO1,2][BL3]**

a) Write a C program to input two unsigned integer numbers. Assume first number as the value on which operation has to be done (say operand). Assume second value as position on which operation will be done(say n). Further program should flip the bit (if it is 0, it should become 1 and if 1 it should become 0) present on the nth position and print the final output. For ex: if operand is 10 and position is 2. Then (10)b = 00000000000000000000000000001010. On flipping the bit position 2 (counting should start from 0), it will become 00000000000000000000000000001110 = 14.

b) Assume you input your roll number to the program and unit digit of your roll number as position value. What will the above program print. For example, if your roll number is 2406189, then first input will be your roll number 2406189 and second input will be the unit digit in your roll number i.e. 9.

Q2. **[5 Points][CO1,2][BL3]**Write a C program to input your roll number and print all digits of it in new lines. For ex: if your roll number is 2406189 it should print each of digit 9 8 1 6 0 4 2 in a new line.

Q3. **[5 Points][CO1,2][BL3]**Write two C programs to take two numbers as input and swap their values. First program should swap the values using a third variables, however second program should do so without any third variable.

Q4. **[1.5+1.5 = 3 Points][CO1][BL2]**

a) Explain the reason behind having mulitple options like int, short, long etc. to store integer data types.

b) Differentiate between signed and unsigned integer.

Q5. **[1+1+1=3 Points][CO1][BL3]**Consider following piece of code as a **complete C program**. Comment in YES/NO if the program will compile and run successfully or not. Write output in case of YES, and explain the reason and correct version of program if NO.

a)
```
int main(){
int a=10,b=20;
return 0;
}
```

b)
```
int main(){
int a=10,b=20;
printf("%d,%d",a,b);
return 0;
}
```

c)
```
int main(){
int a=10,b=20;
printf("%d,%d",a,b);
return 0;
}
```

Q6. **[2+2=4 Points][CO1][BL3]**Guess output of following code snippet assuming there is no error in the program

a)	b)

int a = 10; int b = 40;
a<<b;
printf("%d and %d after %d << %d operation is %d and %d",a,b,a,b,a,b);
	int a = 40;
	int b = 4;
	printf("Output of %d << %d operations is %d",a,b,a<<b);